TODO

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Abstract

Test item retrieval from item banks is an understudied component of exam assembly, particularly in vocational education, where targeted skills, domain-specific terminology, and hierarchical skill relations are not fully captured by distributional text embeddings alone. ESCO provides a structured taxonomy of skills, competences, and occupations. This paper proposes an ESCO-grounded knowledge graph reranker for German vocational education and training (VET) item retrieval. On a German benchmark, the strongest standalone KG reranker, using similarity-weighted direct ESCO links and 512-dimensional SVD representations, improves nDCG@50 by 29.2% over dense retrieval. A late-fusion hybrid with BGEGemma achieves the best overall result, 0.6700 nDCG@50 (+33.8%), and significantly outperforms both components. These results show that ESCO-grounded re-ranking adds complementary skill-aware signals for candidate item retrieval in exam assembly workflows.

1 Introduction

Retrieving test items from an item bank is an important upstream step in exam assembly, especially in vocational education where competency alignment is essential (Linden et al., 2005; Kurdi et al., 2020). The goal here is therefore to improve the candidate pool presented to exam designers, rather than to optimize final test construction under blueprint, fairness, or psychometric constraints. Existing lexical and neural ranking methods do not explicitly model relations among skills and competences. The European Skills, Competences, Qualifications and Occupations (ESCO) framework provides a multilingual representation of such relations and enables ontology-grounded similarity signals for German vocational education and training (VET) item retrieval.[1]

[1]Code and search runs available at: https://anonymous.4open.science/r/ESCO-KGE-reranking-2D1C/

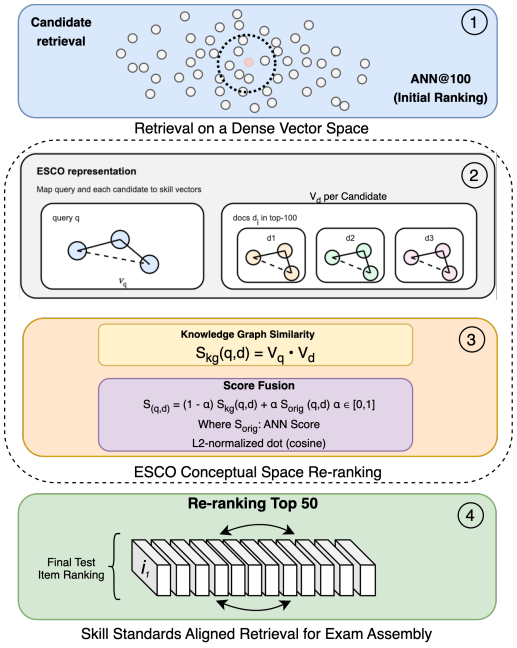

Figure 1: Base re-ranking pipeline: ANN retrieves the top-100 candidate test items; queries and candidates are mapped to ESCO-grounded skill representations to compute $s_{kg}(q, d) = v_q \cdot v_d$, which is fused with s_{orig} via α to re-rank and select the top-50.

Using the benchmark of Palomino et al. (2024), the following research question is addressed: *To what extent can ESCO-based re-ranking improve the relevance of the top-k retrieved test items for German VET item retrieval, compared to lexical baselines and strong text-only re-rankers, including neural and LLM-based models?* (**RQ**). The best standalone variant reaches nDCG@50 = 0.6468, and a late-fusion hybrid with BGEGemma reaches 0.6700. These results indicate that ESCO-derived structured skill signals improve candidate item retrieval for exam assembly workflows and complement strong text-only re-ranking signals. This paper evaluates ESCO-based re-ranking for German VET item retrieval, a setting where ontology-grounded re-ranking has received little prior attention.

2 Related Work

Test item retrieval from item banks is an important but understudied task in educational assessment (Lane et al., 2016; Kurdi et al., 2020). Prior work applies IR methods to distractor suggestion, item-bank management, and assessment pipelines (Ha and Yaneva, 2018; Liu et al., 2005; Zhou et al., 2019; Settles et al., 2020), while recent exam-assembly work ranks items using content, difficulty, and related facets (Palomino et al., 2024, 2025; Do et al., 2025). In parallel, ESCO and related taxonomies have been used in skill-centric NLP for multilingual pretraining, entity linking, skill extraction, framework mapping, and career-resource linking (Zhang et al., 2023, 2024, 2022; Magron et al., 2024; Kavas et al., 2024; Gnehm and Clematide, 2024; Senger et al., 2025). However, ESCO's graph structure has rarely been used as a direct re-ranking signal for German VET item retrieval.

3 Task and dataset

To answer **RQ**, the task is candidate item retrieval: given a query representing the skill set associated with a question in a VET item bank, the goal is to retrieve the most relevant test items from the existing item bank. Because retrieval is the first step in exam construction workflows, this section evaluates whether ESCO-based re-ranking improves top-k retrieval relevance over lexical, neural, and LLM-based baselines.

Testbed. Experiments use the benchmark of Palomino et al. (2024), built from a real German VET item bank used in exam assembly workflows (2,812 test items, 15 German VET queries, and relevance judgments). The benchmark is used to evaluate the proposed re-ranking methods, alternative lexical, neural, and LLM-based baselines, and the late-fusion hybrid extension.

ESCO Taxonomy. ESCO[2] is an EU multilingual taxonomy of skills, competences, qualifications, and occupations. Its hierarchical and associative links provide structured signals for knowledge-informed German VET test item retrieval.

ESCO Knowledge Graph Construction. A multi-relational knowledge graph (KG) is constructed from ESCO (version 23.1), yielding 13,939 entities and 12,530 triples across four relation types. Over the full ESCO skill inventory used as nodes, the resulting relation subset is sparse and long-tailed, with 60.3% isolated nodes and 90.9% of nodes having degree at most 2 (Appendix A.7); construction details are reported in Appendix A.6.

Knowledge Graph Representation Learning. Skill-aware representations are derived using ComplEx (Trouillon et al., 2016), RotatE (Sun et al., 2019), and SVD-based methods. The strongest setting uses similarity-weighted direct ESCO links without graph expansion and 512 latent dimensions; training and representation details are reported in Appendix A.6.

4 Knowledge Graph Based Re-Ranking

To align retrieval with German vocational education and training (VET), the re-ranking approach combines dense retrieval with similarity signals derived from ESCO-based representations. For each query q, a German-language dense retriever[3] implemented with `sentence-transformers` (Reimers and Gurevych, 2020) and `FAISS` (Douze et al., 2024) retrieves a candidate set $\mathcal{C}_q$ of size $K_{cand} = 100$, yielding tuples $(q, d, s_{orig}(q, d))$, where d is a test item and s_{orig} is the original retriever score.

ESCO skill linking. To obtain ESCO-aligned representations, both queries and items are linked to ESCO skills using embedding-based nearest neighbor retrieval over ESCO German skill labels. All ESCO skills are indexed by their `preferredLabel` strings and encoded with the same SentenceTransformer encoder, building a FAISS inner-product index over L2-normalized label embeddings. Given a query text or an item text (stem plus answer options), the top-k nearest ESCO skills are retrieved and only those above a minimum similarity threshold are retained; each retained link is stored as an ESCO `conceptUri` with weight given by the cosine similarity. The linked skill nodes are then mapped to their precomputed ESCO knowledge-graph node embeddings (e.g., SVD/ComplEx/RotatE). In all experiments, queries (top-20) and items (top-15) are linked to ESCO skills with cosine ≥ 0.45, and v_q and v_d are computed as similarity-weighted means of the linked skill embeddings; if no skill is linked, the zero vector is used.

[2] https://esco.ec.europa.eu/en

[3] https://huggingface.co/deutsche-telekom/gbert-large-paraphrase-cosine

The strongest SVD setting uses direct similarity-weighted ESCO links without graph expansion, projected into a 512-dimensional latent space. After row-wise L2 normalization, knowledge graph similarity $s_{kg}(q, d)$ is computed by dot product, and the final score is obtained by convex interpolation:

$$s(q, d) = (1 - \alpha)\, s_{kg}(q, d) + \alpha\, s_{orig}(q, d), \quad (1)$$

where $\alpha \in [0, 1]$ weights s_{kg} vs. s_{orig}. For the strongest standalone SVD setting, a dedicated sweep over $\alpha \in [0.1, 0.9]$ found the best performance in the range $[0.25, 0.35]$, and $\alpha = 0.30$ is therefore reported for this variant. The graph-augmented SVD and KGE-based re-rankers retain $\alpha = 0.5$ as a simple reference setting. For the hybrid system, the best standalone KG re-ranker and BGEGemma are combined over the same dense candidate set after query-wise min-max normalization:

$$s_{hyb}(q, d) = \beta\, \tilde{s}_{svd}(q, d) + (1 - \beta)\, \tilde{s}_{bge}(q, d), \quad (2)$$

where $\tilde{s}_{svd}$ and $\tilde{s}_{bge}$ denote normalized scores and $\beta \in [0, 1]$ controls the contribution of the KG re-ranker. The strongest hybrid setting uses $\beta = 0.55$.

5 Experiments

Thirteen retrieval methods are evaluated, covering ESCO-based, hybrid KG+LLM, lexical, neural, and LLM-based re-ranking approaches. All runs are implemented and evaluated in `PyTerrier` (Macdonald and Tonellotto, 2020). For all re-ranking systems, a top-$K_{cand} = 100$ candidate set is retrieved and the top-$K_{final} = 50$ is evaluated after re-ranking; the ANN baseline selects the top-$K_{final} = 50$ items directly from the dense retriever output. Here, *document* denotes a candidate test item.

Compared Methods:

1. **Hybrid (KG SVD sim-only, 512, $\alpha = 0.30$ + BGEGemma):** A late-fusion hybrid combines the best standalone KG re-ranker with BGEGemma over the shared dense top-100 candidate set using query-wise normalized scores. The reported setting uses $\beta = 0.55$ for the KG component and $(1 - \beta)$ for BGEGemma.
2. **KG re-ranker (SVD sim-only, 512):** Similarity-weighted document-skill and query-skill matrices are built from direct ESCO links without graph expansion; truncated SVD with 512 components is fitted on the document-skill matrix, and latent-space similarity is combined with the original dense score. The reported setting uses $\alpha = 0.30$.
3. **BGEGemma:** An LLM-based re-ranker (Li et al., 2023; Chen et al., 2024)[4] re-scores query-document pairs from the dense top-100 candidate set.
4. **KG re-ranker (SVD graph-augmented, $\alpha = 0.5$):** Sparse document-skill and query-skill matrices are built from KG-augmented ESCO links; truncated SVD is fitted on the document-skill matrix, and latent-space similarity is combined with the original dense score.
5. **KG re-ranker (ComplEx):** Uses ComplEx representations of the ESCO skill graph; query and document vectors are formed by aggregating linked skill embeddings, and their similarity is combined with the original dense score.
6. **KG re-ranker (RotatE):** Uses RotatE representations of the ESCO skill graph; query and document vectors are formed by aggregating linked skill embeddings, and their similarity is combined with the original dense score.
7. **ANN@100→@50:** The top-50 items are selected directly from the dense retriever's top-100 output without re-ranking.
8. **Cross-encoder:** A German cross-encoder[5] re-scores query-document pairs and re-ranks the dense candidates.
9. **Multilingual learned sparse retriever (MLSR):** Because standard sparse baselines such as SPLADE are primarily English/MS MARCO-trained, MLSR serves as the German-compatible sparse neural baseline (Geng et al., 2024)[6]. It re-scores dense candidates using sparse dot-product similarity over lexical-semantic representations.
10. **DPH:** A Divergence From Randomness model, re-ranks candidates by measuring deviation from a random term distribution.
11. **TF-IDF:** Re-ranks the dense candidates by term-overlap similarity weighted by how specific each term is in the corpus.
12. **PL2:** PL2, a DFR-based model with Poisson

[4] https://huggingface.co/BAAI/bge-reranker-v2-gemma
[5] https://huggingface.co/cross-encoder/msmarco-MiniLM-L6-en-de-v1
[6] https://huggingface.co/opensearch-project/opensearch-neural-sparse-encoding-multilingual-v1

#	Method	nDCG	Δ	%	MRR	P	R	F1	MAP
1	**Hybrid (KG SVD sim-only, 512, $\alpha = 0.30$ + BGEGemma)**	**0.6700**	**+0.1694**	**+33.8**	**0.9407**	**0.5760**	**0.4925**	**0.5310**	**0.4349**
2	KG re-ranker (SVD sim-only, 512, $\alpha = 0.30$)	0.6468	+0.1462	+29.2	0.9042	0.5613	0.4814	0.5183	0.4105
3	BGEGemma	0.6008	+0.1002	+20.0	0.9111	0.5187	0.4521	0.4831	0.3667
4	KG re-ranker (SVD graph-augmented, $\alpha = 0.5$)	0.5986	+0.0980	+19.6	0.9200	0.5267	0.4590	0.4905	0.3504
5	KG re-ranker (ComplEx, $\alpha = 0.5$)	0.5516	+0.0510	+10.2	0.8222	0.4787	0.4247	0.4501	0.3083
6	KG re-ranker (RotatE, $\alpha = 0.5$)	0.5497	+0.0491	+9.8	0.8000	0.4773	0.4239	0.4490	0.3058
7	ANN@100→@50	0.5006	+0.0000	+0.0	0.7944	0.4627	0.4113	0.4355	0.2734
8	Cross-encoder	0.4650	-0.0356	-7.1	0.6811	0.4507	0.3973	0.4223	0.2367
9	MLSR	0.4470	-0.0536	-10.7	0.6861	0.4360	0.3797	0.4059	0.2097
10	DPH	0.3625	-0.1381	-27.6	0.7092	0.3320	0.2148	0.2608	0.1301
11	TF-IDF	0.3570	-0.1436	-28.7	0.6978	0.3293	0.2121	0.2580	0.1240
12	PL2	0.3538	-0.1468	-29.3	0.7028	0.3273	0.2100	0.2558	0.1232
13	BM25	0.3179	-0.1827	-36.5	0.6656	0.3107	0.1912	0.2367	0.1066

Table 1: Retrieval results at cutoff 50. Δ and % denote absolute and relative changes in nDCG@50 compared to the dense baseline (ANN@100→@50).

approximation and Laplace smoothing, re-ranks candidates using lexical relevance scores.

13. **BM25:** Re-ranks the dense top-100 candidates with a standard term-matching score that rewards informative query terms.

5.1 Results

Table 1 reports retrieval performance at cutoff 50. The strongest overall method is the Hybrid (KG SVD sim-only, 512, $\alpha = 0.30$ + BGEGemma), which reaches nDCG@50 = 0.6700 (Δ=+0.1694, +33.8%) and yields the largest MAP@50 gain (0.4349 vs. 0.2734). Among standalone methods, KG re-ranker (SVD sim-only, 512, $\alpha = 0.30$) is best at 0.6468 (Δ=+0.1462, +29.2%), followed by BGEGemma at 0.6008 and the graph-augmented SVD setting at 0.5986; ComplEx and RotatE reach 0.5516 and 0.5497. Among the remaining text-only neural baselines, the cross-encoder reaches 0.4650 and MLSR 0.4470, while DPH remains the strongest lexical re-ranker at 0.3625. The stronger sim-only@512 standalone result suggests that higher-rank latent compression of similarity-weighted skill assignments is more beneficial here than one-hop graph expansion, showing that graph expansion is not necessary for the best standalone KG result and reinforcing the view that SVD benefits from modeling global document-skill co-occurrence over a sparse, long-tail ESCO graph (Appendix A.7). The hybrid further indicates that ESCO-grounded and LLM-based re-ranking signals are complementary rather than redundant: it significantly outperforms both the best standalone KG re-ranker and BGEGemma on per-query nDCG@50 (Appendix A.1). A focused ablation over SVD rank and weighting is reported in Appendix A.3; this ablation is conducted under a fixed-α setting, while the standalone headline result additionally benefits from the later $\alpha = 0.30$ tuning.

To test whether the gains of the ESCO-grounded component could be driven by unstable ESCO linking, the linking stage was analyzed under the shared ESCO linking configuration. Linking covers 100.0% of queries and 99.6% of documents, with no zero-link queries and only 12 zero-link documents in the full collection, indicating that the KG-driven gains are not explained by widespread linking failures. Appendix A.2 reports the full diagnostics.

6 Conclusions

To address **RQ**, lexical baselines, neural baselines, ESCO-based re-ranking strategies, and a KG+LLM hybrid re-ranker were compared for German VET item retrieval. Unlike text-only baselines, the proposed approach uses structured ESCO skill signals to compute ontology-grounded similarity without manual skill annotation. All ESCO-based variants improve over the dense baseline and outperform the lexical re-rankers; the strongest standalone variant, KG re-ranker (SVD sim-only, 512, $\alpha = 0.30$), raises nDCG@50 from 0.5006 to 0.6468. A simple late-fusion hybrid with BGEGemma further raises nDCG@50 to 0.6700 and significantly outperforms both the standalone KG re-ranker and BGEGemma, indicating that ESCO-grounded and LLM-based re-ranking signals are complementary. Overall, the results indicate that ESCO-grounded re-ranking can improve candidate item retrieval for exam assembly by complementing dense retrieval with structured skill-aware signals.

7 Limitations and Ethics Statement

This work addresses candidate item retrieval as an upstream support step in exam assembly, improving access to existing items rather than optimizing final test construction or generating new test items. As with any retrieval system, outputs may reflect biases present in the underlying item bank, ESCO coverage, or curricular conventions. The approach is intended to complement standard review procedures: domain experts can use it to surface candidate items more consistently while retaining control over final item selection, skill coverage, and fairness-related decisions.

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A Appendix

A.1 Significance tests

Per-query nDCG@50 differences over 15 queries are tested using paired randomization (sign-flip), a paired t-test, and the Wilcoxon signed-rank test. Table 2 reports p values and the mean difference Δ (system minus baseline) for the strongest standalone KG re-ranker and for the hybrid KG+LLM re-ranker.

A.2 Linking diagnostics

Unit	Top-k	Min sim.	Cov. (%)	Zero	Full (%)	Avg.	Med.	p10	p90
Query	20	0.45	100.0	0	100.0	20.0	0.713	0.642	0.815
Document	15	0.45	99.6	12	94.2	14.6	0.608	0.517	0.713

Table 3: Diagnostics for the ESCO skill-linking stage under the shared ESCO linking configuration. Cov. denotes coverage, Zero the number of zero-link cases, Full the percentage retaining the full top-k, Avg. the average retained links, and Med. the median similarity.

To further examine the behavior of the linking stage, Table 3 reports coverage, zero-link frequency, retained-link counts, and similarity-score distributions under the shared ESCO linking configuration. ESCO linking covers 100.0% of queries and 99.6% of documents, with no zero-link queries and only 12 zero-link documents in the full collection. The current threshold is also permissive rather than brittle: 100.0% of queries retain the full top-20 skills, and 94.2% of documents retain the full top-15, with documents keeping 14.6 links on average. Similarity scores remain consistently above the threshold, with median cosine similarity of 0.713 for queries and 0.608 for documents. Although gold skill annotations are not available to compute linking precision and recall directly, these diagnostics provide complementary evidence that the ESCO linking stage is high-coverage and that the retrieval gains are not driven by frequent linking failures.

A.3 SVD rank and weighting ablation

Variant	Weighting	Graph	Rank	nDCG@50	Δ vs ANN
Graph-augmented ref.	Similarity	1-hop	256	0.5986	+0.0980
Best sim+graph	Similarity	1-hop	384	0.6397	+0.1391
Best sim-only	Similarity	None	512	0.6401	+0.1395
Best binary+graph	Binary	1-hop	384	0.6147	+0.1141
Best binary-only	Binary	None	512	0.6368	+0.1362

Table 4: Focused ablation over SVD rank and skill-assignment weighting in the dense@100→rerank→@50 setup under a fixed interpolation weight $\alpha = 0.5$. The former graph-augmented SVD configuration is included for reference; the remaining rows report the best variant within each weighting family. All best-per-family variants exceed the ComplEx (0.5516) and RotatE (0.5497) reference results.

Comparison	Δ	Rand. p	t p	W p
KG SVD sim-only (512, $\alpha = 0.30$) vs ANN@100→@50	+0.1462	0.00035	0.00007	0.00031
KG SVD sim-only (512, $\alpha = 0.30$) vs Cross-encoder	+0.1817	0.00025	0.00015	0.00031
KG SVD sim-only (512, $\alpha = 0.30$) vs BGEGemma	+0.0469	0.07300	0.07185	0.10942
Hybrid (KG+BGEGemma) vs KG SVD sim-only (512, $\alpha = 0.30$)	+0.0232	0.01860	0.02659	0.02194
Hybrid (KG+BGEGemma) vs BGEGemma	+0.0701	0.00275	0.00309	0.00429
Hybrid (KG+BGEGemma) vs ANN@100→@50	+0.1694	0.00030	0.00002	0.00031
Hybrid (KG+BGEGemma) vs Cross-encoder	+0.2048	0.00005	0.00001	0.00006

Table 2: Paired significance tests on per-query nDCG@50 over 15 queries. Δ is the mean nDCG@50 difference (system minus baseline). Rand. p is a paired randomization (sign-flip) test; t p is a paired t-test; W p is the Wilcoxon signed-rank test. The standalone SVD rows correspond to the tuned $\alpha = 0.30$ setting reported in the main paper.

To further probe why SVD outperforms ComplEx and RotatE, Table 4 reports a focused ablation over latent rank and skill-assignment weighting under a fixed $\alpha = 0.5$. First, the former graph-augmented configuration (similarity-weighted links with 1-hop expansion at rank 256) reaches 0.5986 nDCG@50, confirming that the earlier reported system is reproducible within the same sweep. Second, the strongest standalone SVD setting under this fixed-α analysis is the similarity-weighted direct-link variant without graph expansion at rank 512, which reaches 0.6401; the best graph-augmented similarity-weighted variant is nearly as strong at 0.6397. Third, higher latent ranks consistently help, with the strongest results appearing at 384–512 dimensions, while even the best binary variants remain well above the ComplEx and RotatE references. A later interpolation sweep further improves this standalone sim-only variant to 0.6468 at $\alpha = 0.30$, which is the setting reported in the main paper. Taken together, these results suggest that SVD's advantage is robust across reasonable weighting choices and is driven more by latent compression of the skill-assignment structure than by local graph propagation alone; the main paper's best overall result then comes from late fusion of this strongest standalone KG signal with BGEGemma.

A.6 KG construction and representation details

A multi-relational knowledge graph is constructed from the ESCO taxonomy (version 23.1, released 2023-12-20). Each node represents a skill concept identified by a unique URI; human-readable labels (the *preferredLabel* property) are stored as metadata for indexing and are not part of the graph structure. Edges are derived from two ESCO resources: the *skills hierarchy*, which provides directed edges (*broader*, *narrower*) between adjacent taxonomy levels (Levels 0–3), and the *skill-skill relations* file, which provides non-taxonomic links represented as bidirectional edges (*optional*, *optional_rev*) between related skills. For downstream use, the graph is stored both as typed triples (*head*, *relation*, *tail*) for embedding-based methods and as an undirected adjacency map for neighborhood expansion with exponential-decay weighting.

Skill-aware representations are derived using three ESCO-based methods. ComplEx and RotatE are trained on ESCO triples with `PyKEEN` (Ali et al., 2021) under a standard link-prediction objective, using 512 dimensions for ComplEx and 256 for RotatE; complex-valued embeddings are converted to real vectors by concatenating real and imaginary parts. For the SVD family, sparse document-skill and query-skill matrices are built from ESCO-linked skills, with both graph-augmented and direct-link variants. Truncated SVD is fitted on the document-skill matrix to obtain a shared latent skill space, and queries are projected with the same transformation. Unlike ComplEx and RotatE, which learn directly from ESCO triples, SVD factorizes structured skill assignments; the strongest direct-link setting uses similarity-weighted ESCO links without graph expansion and 512 latent dimensions, explaining 77% of the variance. For ComplEx and RotatE, query and document vectors are computed as confidence-weighted averages of linked skill embeddings. KGE-based vectors use power normalization ($p = 0.5$) followed by L2 normalization, while SVD-based vectors use L2-normalized projections.

A.7 ESCO graph sparsity profile

Statistic	Value
Isolated nodes	8,406 (60.3%)
Leaf nodes (degree = 1)	2,971 (21.3%)
Nodes with degree ≤ 2	12,668 (90.9%)
Median degree	0
90th percentile degree	2
99th percentile degree	7
Maximum degree	25
Largest connected component	2,478 nodes (17.8%)

Table 5: Degree statistics for the 13,939-node, 12,530-triple ESCO graph used in the experiments.

Table 5 makes the graph profile explicit. These percentages are computed over the full ESCO skill inventory used as nodes, while edges are restricted to the relation subset used in the experiments (*broader*, *narrower*, *optional*, *optional_rev*); many skills therefore remain isolated in this task-specific graph rather than indicating missing data. Under this construction, the ESCO graph is highly sparse: 60.3% of nodes are isolated, 21.3% are leaves, and 90.9% have degree at most 2. Its degree distribution is also long-tailed, with median degree 0, 90th percentile degree 2, 99th percentile degree 7, and maximum degree 25; the largest connected component covers only 17.8% of nodes. In addition, the graph is dominated by optional relations (11,258 of 12,530 triples), with only 1,272 hierarchy triples. This profile supports the interpretation that local link-prediction KGEs are constrained by sparse neighborhoods, whereas SVD can benefit from the global co-occurrence structure induced by ESCO-grounded skill assignments.